

Review of Gregory Clark, *To Have and Have Not: The Determinants of Social Status in England, 1600–2026*

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1 Introduction

2 The Fisher/Goldberger Critique

The book’s central empirical exercise is to fit Ronald Fisher’s (1918) classical model of phenotypic assortative mating to the observed pattern of status correlations among relatives. It is worth being precise about what that model assumes, what it implies, and what those implications cost the author, because the costs are severe enough to undercut the book’s central causal claim on its own terms — using only the numbers Clark himself reports.

2.1 The model

Normalize each individual’s status phenotype y to have unit variance, and write it as the sum of an additive genetic component and an environmental component,

$$y = G + E, \quad \text{Var}(G) = h^2, \quad \text{Var}(E) = 1 - h^2, \quad \text{Cov}(G, E) = 0, \quad (1)$$

where h^2 is (narrow-sense) heritability and G, E are independent within a person. This is Clark’s simplified, dominance-free version of Fisher’s model; it is the version whose moment conditions generate the book’s Table A1 and the decay curve $\rho_n = h^2 \left(\frac{1+m}{2}\right)^n$ fit to collateral relatives n steps removed.

Mating is assortative on the *phenotype*: spouses’ y values have correlation

$$\text{Corr}(y_h, y_w) = r, \quad (2)$$

where h, w index husband and wife. Nothing is assumed about how spouses come to resemble each other on y — the model is agnostic about mechanism — but everything that follows is a mechanical consequence of assortment operating on the *whole* phenotype rather than on genotype alone.

2.2 What phenotypic assortment implies for spouses' genes and environments

Because G and E are the only two components of y , and because y is what spouses match on, both components are dragged into correlation across spouses — not just the genetic one. To see this, regress each spouse's genetic component on their own phenotype. Since $\text{Cov}(G, y) = \text{Var}(G) = h^2$ and $\text{Var}(y) = 1$, the regression coefficient is h^2 itself:

$$G_h = h^2 y_h + u_h, \quad \text{Var}(u_h) = h^2(1 - h^2), \quad (3)$$

with u_h uncorrelated with y_h by construction, and an identical equation for the wife. Maintaining the assumption that every cross-spouse correlation runs *only* through the phenotype-matching channel — i.e. that the residuals u_h, u_w are uncorrelated with each other and with the spouse's phenotype — gives

$$\text{Cov}(G_h, G_w) = h^4 \text{Cov}(y_h, y_w) = h^4 r, \quad \text{so} \quad \text{Corr}(G_h, G_w) = h^2 r \equiv m. \quad (4)$$

This recovers Clark's identity $m = h^2 r$: the genotypic correlation between spouses is not a free parameter, it is a forced arithmetic consequence of assorting on the phenotype with correlation r and heritability h^2 .

The same regression logic, applied to E instead of G (regression coefficient $1 - h^2$ rather than h^2 , since $\text{Cov}(E, y) = \text{Var}(E) = 1 - h^2$), gives

$$\text{Cov}(E_h, E_w) = (1 - h^2)^2 r, \quad \text{Cov}(G_h, E_w) = h^2(1 - h^2)r. \quad (5)$$

This is the point Arthur Goldberger raised about Fisher's model, and which I raised independently in my own referee report on the 2023 PNAS paper: phenotypic assortment does not merely correlate spouses' genes, it *also* correlates their environments, and it correlates one spouse's genes with the other's environment. Every other pair of relatives in the model — siblings, cousins, a person and their own best man — is assumed to have *zero* environmental resemblance and zero gene-environment cross-correlation. There is no substantive theory in the book of why marriage, specifically, is the one relationship that generates environmental resemblance while every other family tie generates none. As I put it in my own correspondence on this paper: it is hard to imagine a realistic real-world process that produces environmental resemblance between spouses but not between siblings.

This is not a decorative objection. It is doing real work, as the next two derivations show.

2.3 Transmission, and the sibling correlation

A child's genotype is the average of the two parents' genotypes plus an independent Mendelian segregation term s ,

$$G_{\text{child}} = \frac{1}{2}(G_m + G_f) + s, \quad s \perp G_m, G_f, E_{\text{anyone}}, \quad (6)$$

with $\text{Var}(s)$ pinned down by the equilibrium requirement $\text{Var}(G_{\text{child}}) = h^2$. Two siblings share both parents but draw independent segregation terms s_1, s_2 . Under the model's blanket assumption that non-spousal relatives share no environmental correlation and no gene-environment cross-correlation, the sibling phenotypic covariance reduces to the genetic term

alone:

$$\text{Cov}(y_1, y_2) = \text{Cov}(G_1, G_2) = \text{Var}\left(\frac{1}{2}(G_m + G_f)\right) = \frac{1}{4}[h^2 + h^2 + 2mh^2] = \frac{1}{2}h^2(1 + m). \quad (7)$$

This is the familiar formula behind the collateral-relative decay curve: sibling resemblance runs entirely through the genotypic spousal correlation m .

2.4 The parent-offspring correlation

Now compute the covariance between a child and one parent, say the father. Because the child's own environment E_{child} is assumed independent of everything about the parents,

$$\text{Cov}(y_{\text{child}}, y_f) = \text{Cov}(G_{\text{child}}, G_f) + \text{Cov}(G_{\text{child}}, E_f). \quad (8)$$

The first term is exactly the genetic term computed above for siblings,

$$\text{Cov}(G_{\text{child}}, G_f) = \frac{1}{2}[\text{Cov}(G_m, G_f) + \text{Var}(G_f)] = \frac{1}{2}h^2(1 + m). \quad (9)$$

But there is a second term that siblings do not have: the child inherits half its genes from the mother, and the mother's genes are cross-correlated with the *father's environment* through the marriage — exactly the term derived in equation (5):

$$\text{Cov}(G_{\text{child}}, E_f) = \frac{1}{2} \text{Cov}(G_m, E_f) = \frac{1}{2}h^2(1 - h^2)r. \quad (10)$$

Adding the two terms and substituting $m = h^2r$:

$$\begin{aligned} \text{Cov}(y_{\text{child}}, y_f) &= \frac{1}{2}h^2(1 + m) + \frac{1}{2}h^2(1 - h^2)r \\ &= \frac{1}{2}h^2[1 + h^2r + (1 - h^2)r] \\ &= \frac{1}{2}h^2[1 + r]. \end{aligned} \quad (11)$$

So $\rho_{PO} = h^2(1 + r)/2$: the parent-offspring correlation runs through r , not m . This is worth pausing on. It is precisely the spousal gene-environment cross-term that Goldberger flagged as implausible — the term with no analogue anywhere else in the model — that is mathematically responsible for parent-offspring resemblance exceeding sibling resemblance. The "curious exception" for spouses is not a side remark; it is load-bearing.

2.5 The decisive test: ρ_{PO} vs. ρ_S

Comparing equations (7) and (11), and using $m = h^2r$ again:

$$\rho_{PO} - \rho_S = \frac{1}{2}h^2(r - m) = \frac{1}{2}h^2r(1 - h^2) > 0 \quad \text{whenever } 0 < h^2 < 1. \quad (12)$$

The model does not merely permit parent-offspring correlation to exceed sibling correlation — it *requires* it, by a specific, computable amount, given only the two parameters Clark himself estimates.

Plug in Clark’s own reported values, $r \approx 0.8$ and $m \approx 0.6$, which imply $h^2 = m/r = 0.75$. Equation (12) predicts

$$\rho_{PO} - \rho_S \approx \frac{1}{2}(0.75)(0.8 - 0.6) = 0.075. \quad (13)$$

The gap Clark’s own data actually show is smaller than this by roughly two orders of magnitude. This is not a subtle specification-testing exercise requiring auxiliary assumptions — it is arithmetic, performed entirely with numbers the author reports himself. Clark comes close to seeing the problem (p. 61: parity between the two correlations “would only be predicted...if parents were actually matching on the genotype itself”) but never computes the size of the discrepancy his own model predicts, and does not revise the model, or his confidence in its fit, in light of it.

2.6 Two parameters, not three: the Jinks-and-Eaves precedent

Equation (4) is the source of a second, closely related problem. Because $m = h^2 r$ is a forced identity rather than an independent empirical fact, the model has exactly two free structural parameters — one may parameterize it as (h^2, r) or as (h^2, m) , but not as all three (h^2, r, m) simultaneously without imposing the constraint that links them.

This is not a new observation; it is the identical error Goldberger (1978) diagnosed in Jinks and Eaves’s reanalysis of the British and American IQ-kinship data. Jinks and Eaves treated the genotypic spousal correlation A as a free parameter, in violation of Fisher’s own requirement that $A = c_1 c_2 m$, and their fitted estimates duly violated it (for the British data set, $c_1 c_2 m = .22$ against a fitted $A = .47$). When Goldberger refit the model with the constraint properly imposed, the fit measurably worsened: χ^2 rose from 9.05 (9 d.f., A free) to 13.65 (10 d.f., constraint imposed) — an increment of 4.60 on one degree of freedom, significant at the 5% level, which Goldberger read as “a piece of evidence against the classical model” even though the constrained model still fit acceptably well in absolute terms.

Clark’s book (Technical Appendix, eqs. A1–A3) fits m from the collateral-relative decay curve and r from marriage records as two essentially separate exercises, and at no point jointly imposes the two-parameter constraint across all relationship types — parent-offspring included — simultaneously. This is the same practice Goldberger’s re-analysis showed to be non-innocuous three decades before the underlying PNAS paper was written, and it is a point I raised directly in my referee report on that paper, where Clark’s response (that spouses may be assorting on a “wider phenotype” than the one being modeled) did not resolve whether two or three parameters were actually free in the estimation. The book inherits this unresolved status unchanged.

3 The Identification Problem

4 What “Genetic” Does Not Mean

5 Conclusion